

S&P 500 Return Forecasting

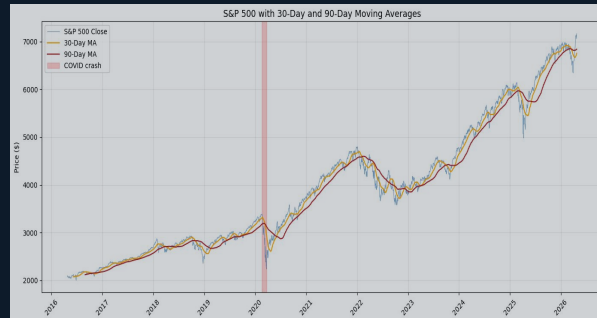
Using Macroeconomic Indicators & Machine Learning

Dataset: S&P 500 daily prices + 4 macro indicators | 2016 – 2026

2,515 trading days | 5 models compared | 32 engineered features

Best Model: XGBoost | RMSE 0.007231 | MAE 0.006751

Hritika Jadeja, Jeff Rathgeber, Ryan Kahrmanian, Xinyao Dong



What We Built & Why



The Question

Can macroeconomic data predict next-day S&P 500 log returns? We test 5 models from simple regression to deep learning.



The Data

Daily S&P 500 prices merged with Fed Funds Rate, GDP (quarterly), Core Inflation, and Unemployment Rate — all aligned to trading days.



The Goal

Build a rigorous ML pipeline: clean → stationarize → engineer features → train → evaluate — and identify which model wins.

Raw Data

Clean & Align

Stationarize

Feature Eng.

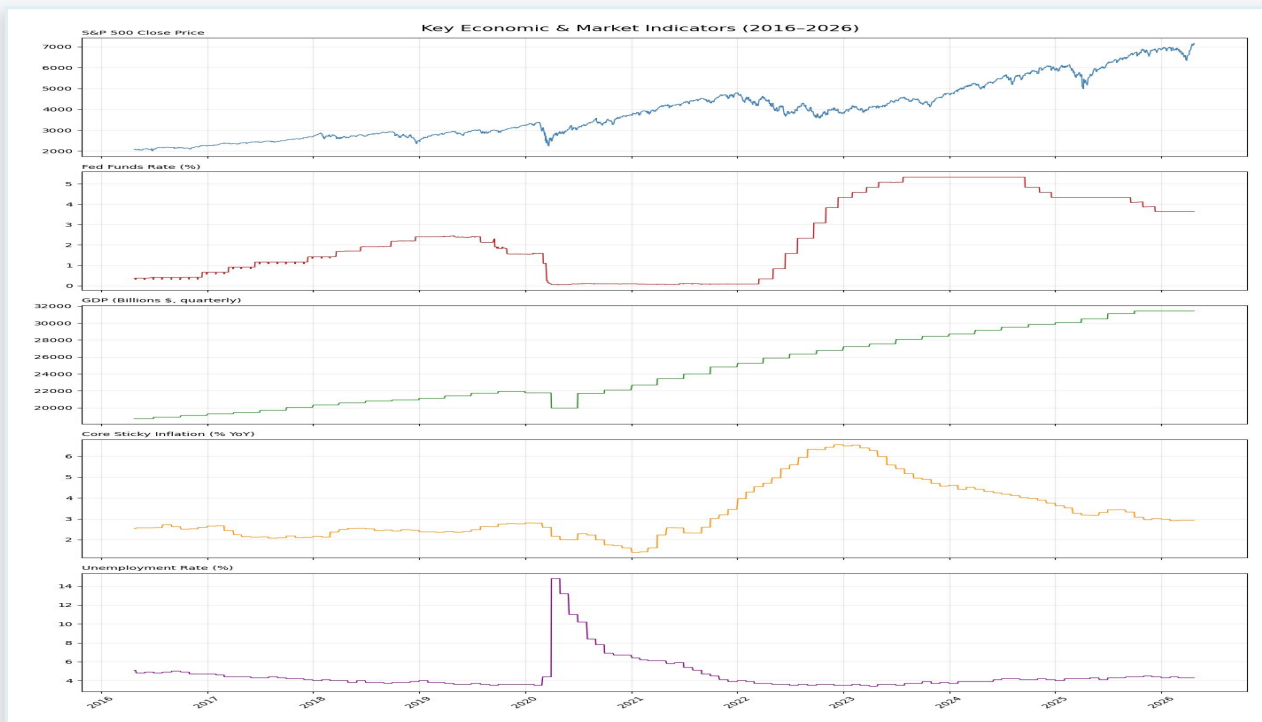
Train Models

Evaluate

End-to-end pipeline from raw financial data to model evaluation

5 Key Economic & Market Indicators (2016 – 2026)

S&P 500 close price, Fed Funds Rate, GDP, Inflation, Unemployment — 2,515 trading days



SPX

2,087 → 6,700+
3x growth over 10 years

Fed Rate

0.37% → 5.33%
Peak hiking cycle 2022–23

GDP

\$18.7T → \$31T
–3.4% COVID dip in 2020

Inflation

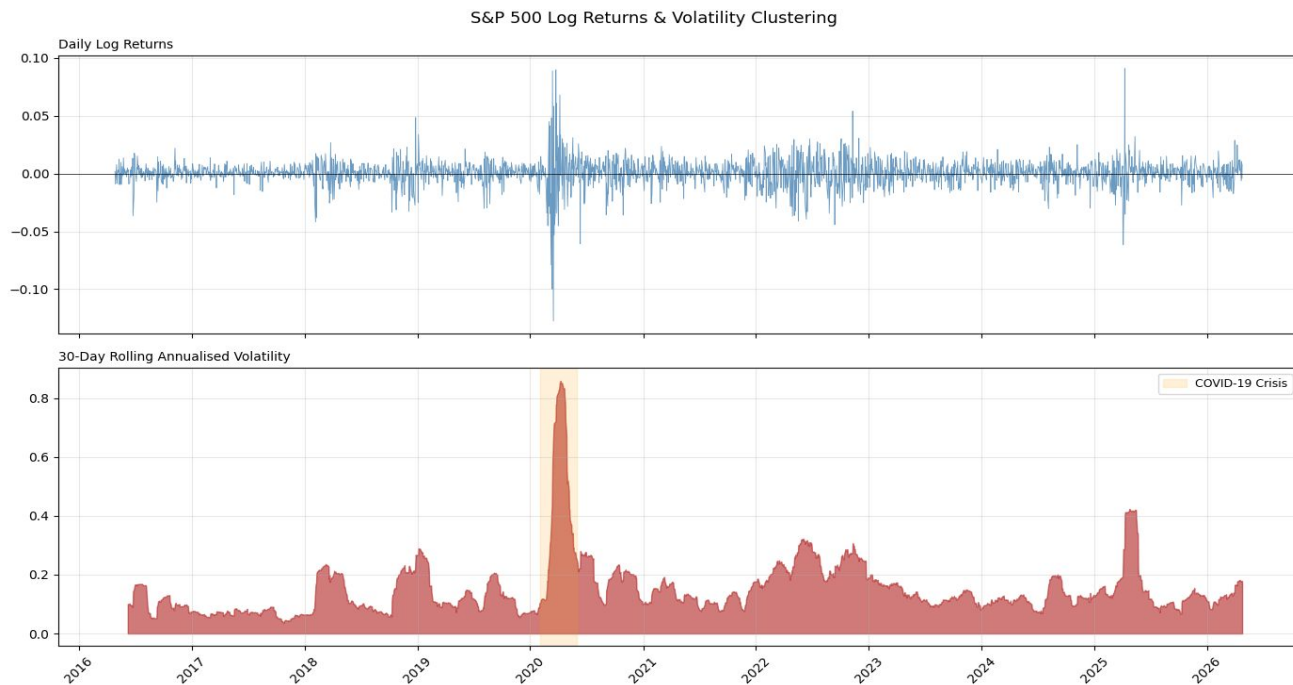
2.5% → 6.8%
Post-COVID surge

Unemployment

5.1% → 14.8%
COVID spike, then recovery

Log Returns & Volatility Clustering

COVID-19 shock visible as extreme spike; 30-day rolling annualised volatility peaked at ~85%



COVID Crash

Daily return hit -12% on Mar 16, 2020 — the worst single day in the dataset. Annualised volatility spiked to 85%.

Clustering

Volatility clusters are clearly visible: calm periods alternate with turbulent regimes (2018, 2020, 2022, 2025).

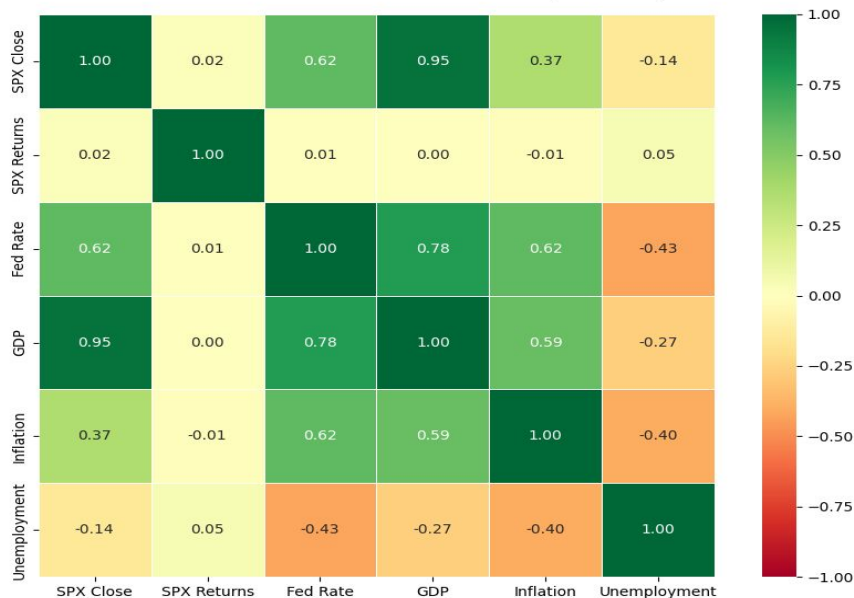
Mean Reversion

Log returns oscillate tightly around zero — a key assumption validating their use as the model target.

Correlation Matrix & Moving Average Signals

Correlation Matrix

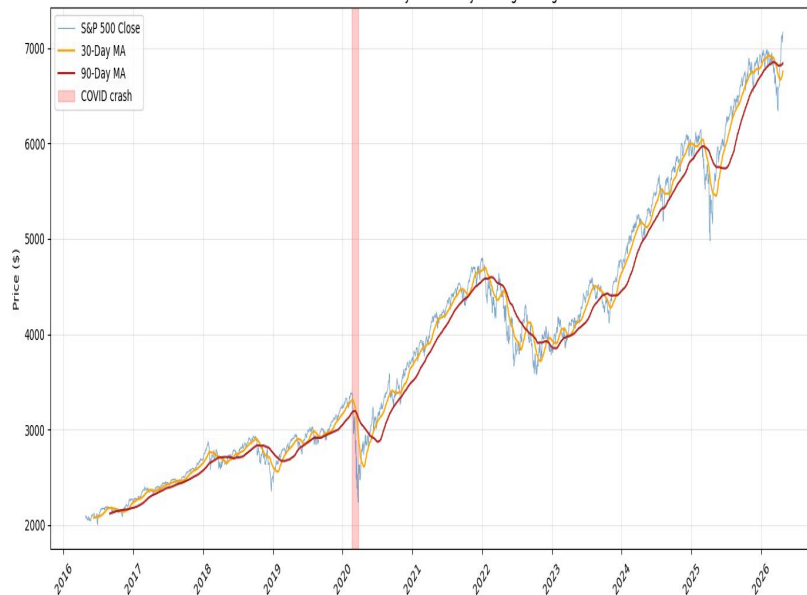
Correlation Matrix: Economic & Market Indicators (2016–2026)



$GDP \times SPX: +0.95$ | $Inflation \times Fed: +0.62$ | $Unemployment \times Fed: -0.43$

30-Day & 90-Day Moving Averages

S&P 500 with 30-Day and 90-Day Moving Averages



Death cross (Mar 2020) confirmed crash | Golden cross (Jul 2020) confirmed recovery

ADF Test — Making the Data Model-Ready

Analogy: Like calibrating GPS — models need changes, not levels. We first-difference every non-stationary series.

Raw Series

Series	ADF Stat	p-val	Result
SPX Close	0.5345	0.9859	NO ✗
Fed Rate	-0.8623	0.8001	NO ✗
GDP	0.2086	0.9727	NO ✗
Inflation	-1.4616	0.5524	NO ✗
Unemployment	-3.4138	0.0105	YES ✓
Log Return	-15.9007	0.0000	YES ✓

After First-Diff

Series	ADF Stat	p-val	Result
Δ SPX Close	-16.4019	0.0000	YES ✓
Δ Fed Rate	-51.7897	0.0000	YES ✓
Δ GDP	-50.3790	0.0000	YES ✓
Δ Inflation	-5.4043	0.0000	YES ✓
Δ Unemployment	-10.4311	0.0000	YES ✓
Log Return	-15.9007	0.0000	YES ✓

✓ All 6 series confirmed stationary after first-differencing | Stationary master shape: (2,514 × 5) | All series ready for modelling

Building 32 Features Across 3 Categories

Multi-lag features + technical indicators + interaction terms → expanded feature matrix

July 17

Multi-Lag Features

25 features

- ▶ Δ Fed Rate, Δ GDP, Δ Inflation, Δ Unemployment at lags 1, 2, 3, 5, 10
- ▶ Log returns at each lag — captures autocorrelation
- ▶ Like checking yesterday's weather to forecast today



Technical Indicators

4 features

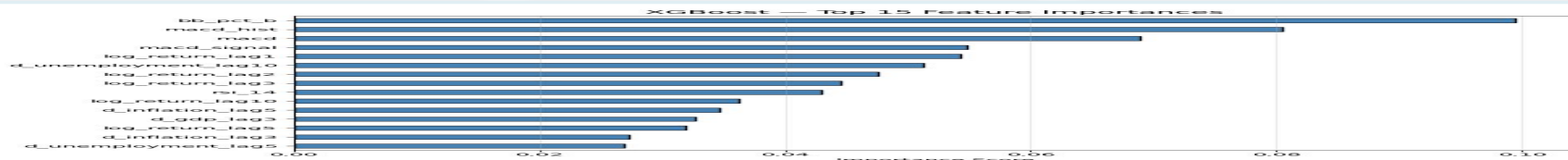
- ▶ RSI (14-day): overbought/oversold momentum signal
- ▶ MACD (12/26 EMA + signal line): trend direction
- ▶ Bollinger Band %B (20-day): price vs. volatility range



Interaction Terms

3 features

- ▶ Fed Rate $\times$ Inflation: combined tightening pressure
- ▶ GDP $\times$ Unemployment: growth vs. labor market health
- ▶ XGBoost top feature: Bollinger %B (importance 0.10)



5 Models: From Baseline to Deep Learning

All trained on 80% (2,010 rows) and evaluated on 20% test set (Apr 2024 – Apr 2026)

01 Linear Regression [Baseline]

Weighted sum of lagged macro features.
Sets performance floor — if nothing beats this, macro lags alone have weak predictive power.

02 Random Forest [Ensemble]

500 trees on original 4-feature set.
Captures non-linear regime shifts, e.g. rate hikes during high inflation.

03 VAR [Time Series]

Vector Autoregression (lag=1 by AIC).
Captures feedback loops — how an unemployment spike feeds back into returns over days.

04 LSTM [Deep Learning]

2-layer LSTM (seq=30, 64 units). Trained with early stopping. Captures 2020-style volatility clusters via sequence memory.

05 XGBoost [Boosting ✓]

500 estimators, lr=0.01, depth=4. Trained on 32-feature set. Best performer: RMSE 0.007231, top signals were price-based technicals.

Actual vs Predicted Log Returns — All 5 Models

Test period: Apr 2024 – Apr 2026 | Each panel = one model | XGBoost (bottom) shows closest prediction tracking



LR: 0.01024

RF: 0.01024

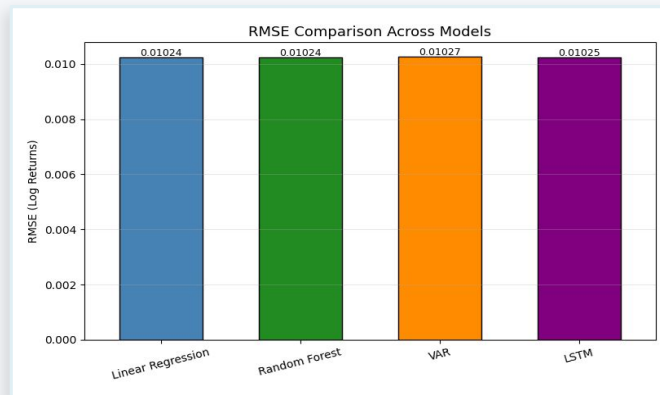
VAR: 0.01027

LSTM: 0.01025

XGBoost: 0.00723 🏆

Model Performance — Test Period: Apr 2024 – Apr 2026

Model	MAE	RMSE	MAPE (%)	Verdict
Linear Regression	0.006775	0.010242	169.24	Baseline
Random Forest	0.006770	0.010237	171.07	Strong
VAR	0.006859	0.010272	243.99	Weak MAPE
LSTM	0.006763	0.010262	165.56	Near-best
XGBoost ✓	0.006751	0.007231	163.18	 Best



RF wins on base metrics

With 4 macro features: Fed Rate lag dominates at 0.74 importance

XGBoost wins with all features

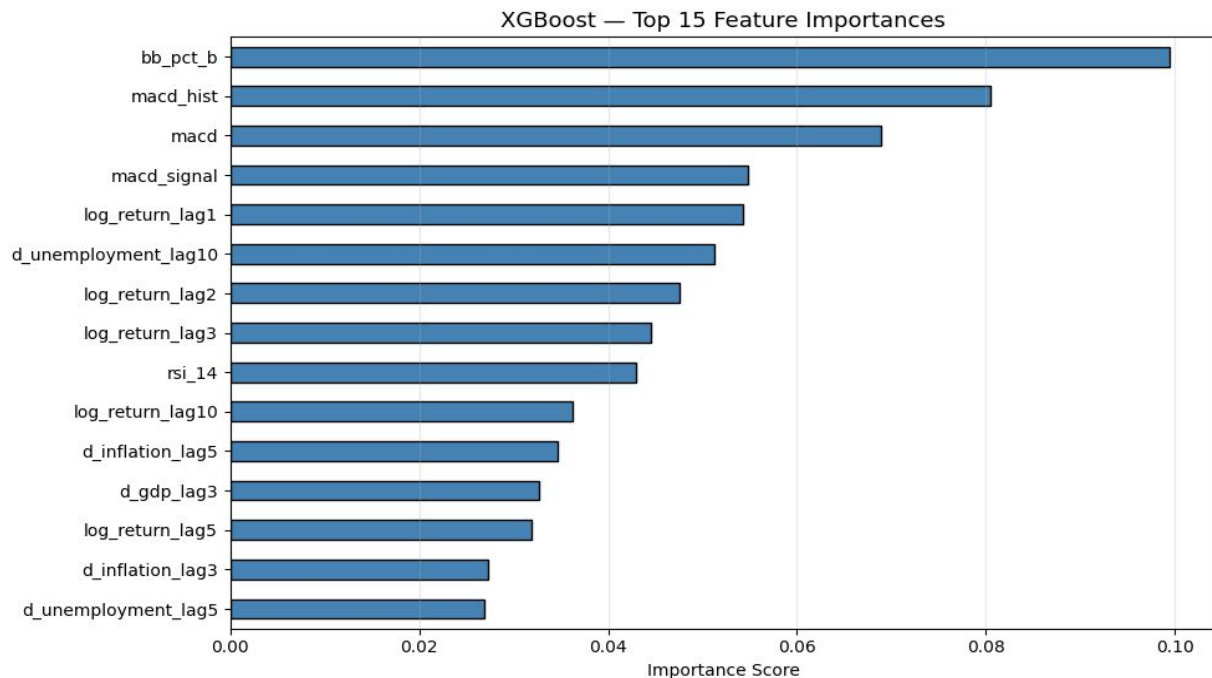
Technical signals (Bollinger %B, MACD) are more predictive than macro lags

MAPE misleading here

Near-zero returns inflate % error; RMSE & MAE are more reliable metrics

XGBoost — Top 15 Feature Importances

Technical price signals dominate over raw macro lags — confirms the value of feature engineering



Bollinger %B

Importance: 0.0995

Price position within volatility band — top signal by far

MACD Histogram

Importance: 0.0805

Momentum shifts captured by MACD cross signals

MACD Line

Importance: 0.0689

Trend direction from exponential moving averages

Log Return Lag1

Importance: 0.0542

Yesterday's return — mild autocorrelation in daily data

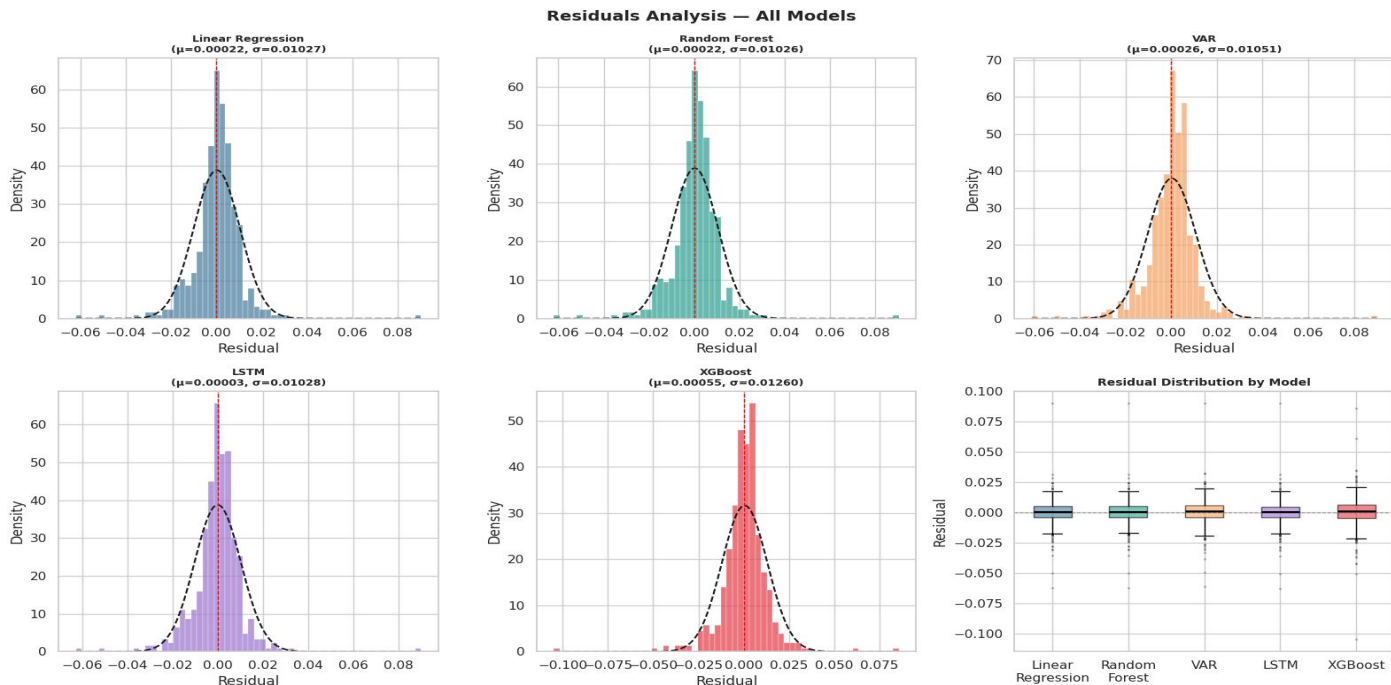
RSI (14-day)

Importance: 0.0430

Momentum indicator; overbought/oversold signal

Residual Distributions — All 5 Models

All models show near-zero mean residuals and heavy tails — classic pattern for financial returns



Lin. Reg

$\mu=0.00022$

$\sigma=0.01027$

Rand. Forest

$\mu=0.00022$

$\sigma=0.01026$

VAR

$\mu=0.00026$

$\sigma=0.01051$

LSTM

$\mu=0.00003$

$\sigma=0.01028$

XGBoost

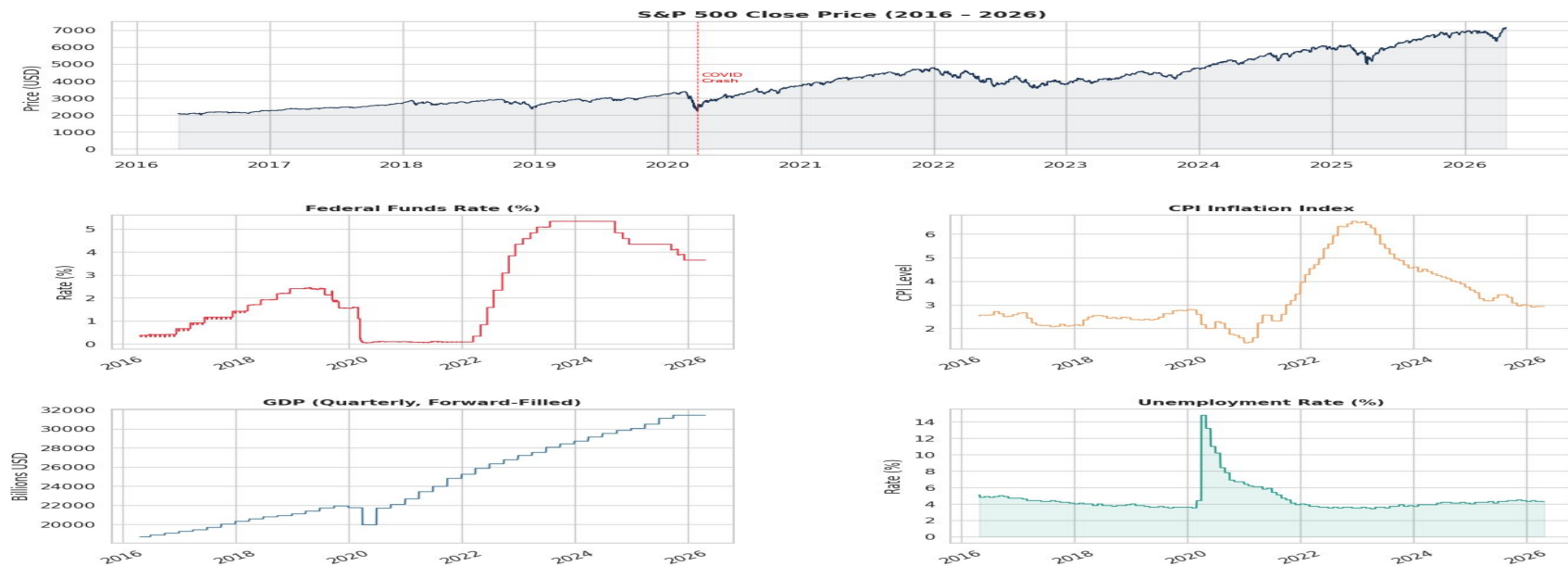
$\mu=0.00055$

$\sigma=0.01260$

Economic Indicators Dashboard

S&P 500 + Fed Rate + Inflation + GDP + Unemployment — annotated with COVID crash marker

Macro Economic Indicators — Full Sample (2016-2026)



COVID crash line (Mar 23, 2020) annotated in red — the single most impactful event in the 10-year dataset

Summary & Key Takeaways

1

Stationarity is non-negotiable All 5 raw macro series were non-stationary. First-differencing made them valid model inputs.

2

Feature engineering drives performance Adding RSI, MACD, Bollinger signals caused XGBoost to outperform. The baseline Linear Regression gap was meaningful.

3

XGBoost wins — but not by much Best RMSE (0.007231) and MAE (0.006751). Other models cluster tightly around 0.01024–0.01027 RMSE.

4

Technical signals > macro lags Top XGBoost features were Bollinger %B, MACD Hist, MACD — price-based, not macro. Macro lags ranked lower.

5

Markets are hard to beat All models show >160% MAPE. Daily log returns are near-zero on average — predicting direction is the real challenge.